Hypothesis two sample t-tests

1

- **a** $p = 0.73 > \alpha$, therefore do not reject the null hypothesis.
- **b** $p = 0.74 > \alpha$, therefore do not reject the null hypothesis.
- **c** $p = 0.14 < \alpha$, therefore reject the null hypothesis.

2

- **a** μ_1 is mean hours of study for Year 11 students.
  μ_2 is mean hours of study for Year 12 students.
  $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 < \mu_2$
- **b** One tailed test
- **c** $p = 0.000342$
- **d** As $p < \alpha$ we say that there is enough evidence to reject H_0 on a 1% level of significance. Therefore we can comment that Year 11 students do appear to study less than Year 12 students.

3

- **a** μ_1 is mean bulb hours for the old globes. μ_2 is mean bulb hours for the new globes.
  $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 < \mu_2$
- **b** One tailed test
- **c** $p = 0.04$
- **d** As $p < \alpha$ we say that there is enough evidence to reject H_0 on a 5% level of significance. Therefore we can comment that the new globes do appear to last longer than the old globes.

4

- **a** The population data is normally distributed and the two sets of data have equal standard deviations (or variances).
- **b** μ_1 is mean deaths using poison A. μ_2 is mean deaths using poison B.
  $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 \neq \mu_2$
- **c** $p = 0.16$
- **d** As $p > \alpha$ we do not reject the null hypothesis. There is no evidence that poison A is better or worse than poison B.

5

- **a** $H_0 : \mu_A = \mu_B, H_1 : \mu_A \neq \mu_B$
- **b** Two tailed test
- **c** $p = 0.077$
- **d** As $p > \alpha$ we do not reject the null hypothesis that the drugs perform the same and there is no statistically significant difference in the sleep outcomes.

6

- **a** $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 \neq \mu_2$
- **b** Two tailed test
- **c** $p = 0.01$
- **d** As $p < \alpha$ we reject the null hypothesis that the average scores are the same for both groups and we can say that there is a statistically significant difference between the test scores of the males compared to the females.
- **e** There is a 10% chance of incorrectly rejecting H_0.

7

- **a** $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 > \mu_2$
- **b** $p = 0.53$ therefore $p > \alpha$ and we do not reject the hypothesis that native speakers score higher than non-native speakers on the reading test.